

PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR

Department: Computer Technology

Semester: VIII Section: A and B

CAT-II (2023-24)

Subject: Blockchain and its Applications (Elective VII)

Subject Code: BECT802T-2

Duration: 1.5 Hrs

Max. Marks: 35

Note:

- 1) All questions are compulsory.
- 2) All questions carry marks as indicated.
- 3) Due credits will be given on neatness.
- 4) Draw diagram wherever it is necessary.

Questions		Marks	CO	BL
Q. 1	A. Can we reach a consensus when there is one commander, one good lieutenant, and one faulty lieutenant in a Byzantine Generals Problem. Yes, or No? a. Yes b. No	1	CO3	1
	B. Which are the properties of an asynchronous consensus: a. Validity b. Agreement c. Termination d. All the above	1	CO3	1
	C. What is Nonce? How is it used to solve puzzles in blockchain?	5	CO3	2
OR				
Q. 2	A. If you have f number of faulty nodes, then you need at least how many replicas to reach consensus irrespective of crash fault or byzantine fault. a. $2f + 1$ b. $3f + 1$ c. $f + 1$ d. $3f$	1	CO3	3
	B. What is the correct sequence of operations in PBFT algorithm i. Prepare ii. Reply iii. Commit iv. Pre-Prepare a. iv, i, ii, iii b. iv, i, iii, i c. i, iv, ii, iii d. I, ii, iv, iii	1	CO3	2
	C. Differentiate Proof of Work and Proof of Stake	5	CO3	3
Q. 3	A. Which of the following is an open, scalable consensus algorithm having low transaction throughput? a. PoW b. PoS c. PBFT d. PoL	1	CO4	1
	B. Hyperledger Fabric only allows Proof of Work consensus to be plugged in to ensure a high degree of trustworthiness. True or False a. True b. False	1	CO4	1
	C. Draw and explain Ethereum Network.	4	CO4	3
	D. Explain the blockchain performance triangle.	8	CO4	3
OR				
Q. 4	A. BFT protocol ensures finality in general. a. False b. True	1	CO4	1
	B. Data transfer is an important aspect of interoperability in which of the following blockchains? a. Hyperledger Indy b. Corda c. Hyperledger Fabric d. None	1	CO4	1
	C. Discuss Ethereum Virtual Machine - EVM	5	CO4	3
	D. What are the function modifiers in solidity? How is blockchain distributed ledger different from the traditional ledger?	7	CO4	2
Q. 5	A. Which of the following is true about Single Sign-on a. The same identity can be used to access multiple services b. Decentralized providers always maintain the identity of individuals participating in single sign-on c. All individuals use a fixed same identity for every one of them d. All identity holders are also identity providers	1	CO5	1
	B. Which of the following is not a valid distinct component in Distributed Identifiers (DID) Architecture. a. DID Controller b. Verifiable Data Registry c. DID Subject d. DID Randomizer	1	CO5	3

	C. What is Identity? Explain Centralized Digital Identity	5	CO5	2
	D. Explain in detail how can blockchain be used for identity management.	7	CO5	2
OR				
Q. 6	A. Which of the following is used as a public permissioned ledger based DID registry? a. Hyperledger Indy b. Bitcoin c. Ethereum d. Sidetree	1	CO5	1
	B. In Verifiable Credential (VC), a claim is a statement about a non participant. a. True b. False	1	CO5	1
	C. Discuss Decentralized Identifiers (DIDs)	5	CO5	2
	D. Discuss problem with interoperability in blockchain using an example	7	CO5	2